

## Pandas Functions Mastery Roadmap for Data Analysts

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### Goal

Master ~38 Pandas functions to cover 90% of real-world data analyst tasks, including IPL dataset practice.

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### LEVEL 1: MUST-KNOW BASICS (Very Easy)

**Functions:** `pd.read_csv()`, `.head()`, `.tail()`, `.info()`, `.shape`, `.columns`, `.describe()`

**IPL Practice:** - Check dataset size - Inspect column types - View sample rows

**Highlight:** Foundational skills for loading and exploring data.

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### LEVEL 2: SELECTION & FILTERING (Easy)

**Functions:** `df['column']`, `.loc[]`, `.iloc[]`, Boolean filtering, `df[df['batsman_runs'] > 4]`

**IPL Practice:** - Find all sixes - Filter by one team - Filter by one player

**Highlight:** Essential for slicing and selecting relevant data.

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### LEVEL 3: GROUPING & AGGREGATION (Medium)

**Functions:** `.groupby()`, `.sum()`, `.mean()`, `.count()`, `.size()`, `.agg()`

**IPL Practice:** - Total runs by team - Strike rate calculation - Matches per season

**Highlight:** Core analytics skills for summarizing data.

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### LEVEL 4: SORTING & RANKING (Medium)

**Functions:** `.sort_values()`, `.head()`, `.tail()`, `.nlargest()`

**IPL Practice:** - Top 10 batters - Best strike rate - Most sixes

**Highlight:** Identify top performers and rank metrics.

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## LEVEL 5: DATA CLEANING (Medium)

**Functions:** `.isnull()`, `.dropna()`, `.fillna()`, `.duplicated()`, `.drop_duplicates()`, `.astype()`

**IPL Practice:** - Handle missing players - Correct data types

**Highlight:** 80% of data analyst work involves cleaning raw data.

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## LEVEL 6: STRING & DATE OPERATIONS (Medium-Hard)

**Functions:** `.str.lower()`, `.str.contains()`, `.str.replace()`,  
`pd.to_datetime()`, `.dt.year`, `.dt.month`

**IPL Practice:** - Clean player names - Season-wise analysis

**Highlight:** Useful for text and time-based data transformations.

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## LEVEL 7: COMBINING DATA (Hard)

**Functions:** `pd.merge()`, `pd.concat()`

**IPL Practice:** - Merge matches & deliveries - Add match context to deliveries

**Highlight:** Combine multiple datasets for richer insights.

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## LEVEL 8: ADVANCED ANALYST TOOLS (Hard)

**Functions:** `pd.pivot_table()`, `.value_counts()`, `.reset_index()`, `.set_index()`, `.apply()`

**IPL Practice:** - Powerplay analysis - Team vs team summaries

**Highlight:** Advanced tools for custom calculations and pivot-style analysis.

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# Industry Truth

Master these ~38 functions and you can: - Perform 90% of data analyst tasks - Analyze any dataset (not just IPL) - Prepare for interviews confidently

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## Suggested Study Flow

1. Master Level 1 → 2 → 3 sequentially
2. Take small projects on IPL dataset after each level
3. Make notes, practice mini-quizzes, and gradually move to Level 8
4. Combine multiple levels for real analysis projects

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**End of Roadmap**